

# QUADAS-C: A Tool for Assessing Risk of Bias in Comparative Diagnostic Accuracy Studies

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Comparative diagnostic test accuracy studies assess and compare the accuracy of 2 or more tests in the same study. Although these studies have the potential to yield reliable evidence regarding comparative accuracy, shortcomings in the design, conduct, and analysis may bias their results. The currently recommended quality assessment tool for diagnostic test accuracy studies, QUADAS-2 (Quality Assessment of Diagnostic Accuracy Studies-2), is not designed for the assessment of test comparisons. The QUADAS-C (Quality Assessment of Diagnostic Accuracy Studies-Comparative) tool was developed as an extension of QUADAS-2 to assess the risk of bias in comparative diagnostic test accuracy studies. Through a 4-round Delphi study involving 24 international experts in test evaluation and a face-to-face consensus meeting, an initial version of the tool was developed that was revised and finalized following a pilot study among potential users. The QUADAS-C tool retains the same 4-domain structure of QUADAS-2 (Patient Selection, Index Test, Reference Standard, and Flow and Timing) and comprises additional questions to each QUADAS-2 domain. A risk-of-bias judgment for

comparative accuracy requires a risk-of-bias judgment for the accuracy of each test (resulting from QUADAS-2) and additional criteria specific to test comparisons. Examples of such additional criteria include whether participants either received all index tests or were randomly assigned to index tests, and whether index tests were interpreted with blinding to the results of other index tests. The QUADAS-C tool will be useful for systematic reviews of diagnostic test accuracy addressing comparative questions. Furthermore, researchers may use this tool to identify and avoid risk of bias when designing a comparative diagnostic test accuracy study.

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**S**tudies of diagnostic test accuracy (DTA) are pivotal in the evaluation of new and existing diagnostic tests and strategies (1). These studies can not only evaluate the accuracy of a single index test but can also evaluate multiple index tests and compare their accuracy.

Comparison of test accuracy is preferably done in studies directly comparing index tests in the same study, also known as comparative DTA studies (2, 3). Comparative DTA studies have the potential to provide rigorous evaluations of test comparisons, unlike comparisons based on separate studies evaluating the accuracy of single tests (4, 5). However, like any study, comparative DTA studies need to be evaluated for their validity and applicability before their results can be used for guiding health care decisions.

Comparative DTA studies are susceptible to sources of bias that are not captured by the recommended QUADAS-2 (Quality Assessment of Diagnostic Accuracy Studies-2) tool for assessing the methodological quality of DTA studies (6). In test comparisons, bias may arise, for instance, when participants receiving index test A represent a different disease spectrum than those receiving index test B, or when results of A are interpreted with knowledge of the results of B and vice versa (7). To account for these and other potential sources of bias,

risk-of-bias assessments need to include additional items specific to comparisons of test accuracy.

An overview of 238 comparative DTA systematic reviews that were published in 2017 showed that risk-of-bias assessments for test comparisons had been planned or conducted in only 2 reviews (3). Furthermore, the overview did not identify any risk-of-bias tools designed for comparative DTA studies.

We developed QUADAS-C (Quality Assessment of Diagnostic Accuracy Studies-Comparative), a tool for assessing risk of bias in comparative DTA studies. The QUADAS-C tool is not designed as a standalone tool but as an extension of QUADAS-2. Although QUADAS-C is designed for use in systematic reviews, investigators can also consult it during the planning and design phases of a comparative DTA study to reduce risk of bias. In this article, we explain the development process of QUADAS-C, its scope, and how it should be used.

## COMPARATIVE ACCURACY QUESTIONS

We first briefly explain what comparative accuracy questions are and how they differ from questions regarding single-test accuracy (Table 1 outlines key differences). Comparative accuracy questions ask how the accuracy of an index test compares with that of another index test for detecting the same target condition. For example, one such question would be whether Xpert MTB/RIF Ultra (Cepheid) is more sensitive for diagnosing tuberculous meningitis than Xpert MTB/RIF (8). For a valid comparison, participants receiving index test A should be exchangeable with participants receiving

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index test B. This can be accomplished by each participant undergoing all index tests (often called a fully paired or within-subject design) or approximated by randomly allocating participants to index tests (randomized design) (2, 9, 10). Comparative accuracy results can be expressed as absolute or relative differences in sensitivity and specificity, predictive values, area under the curve, or other measures of accuracy, including decision analytic measures such as net benefit (11, 12). Knowledge of comparative accuracy is important for the selection and recommendation of a test from many alternative, competing tests, especially when studies evaluating the effectiveness of test-treatment strategies on patient-important outcomes are absent (13). A key characteristic of comparative accuracy questions is that none of the tests being compared is the reference standard. Rather, the reference standard is a means to verify whether participants have the target condition.

### DEVELOPMENT OF QUADAS-C

The process of developing QUADAS-C was based on a framework for developing quality assessment tools by Whiting and colleagues (14). A steering group consisting of 8 people with a background in diagnostic test evaluation and/or systematic review methods coordinated all activities. Six members of the steering group (P.F.W., S.M., J.J.D., M.M.G.L., C.F.D., and C.J.H.) had also been involved in the development of QUADAS-2.

### Delphi Study

For achieving consensus on the scope of the tool and which items to include in it, we conducted a Delphi study (protocol registered at <https://osf.io/tmze9>). The study was designed as 4 rounds of surveys interspersed with feedback of results to all panel members. After each round, the steering group held a teleconference to discuss the results of the previous round and the design of the next round.

We invited international experts in the field of DTA research to participate in the study; they were identified based on the recommendations of individual steering group members. The 16 experts who accepted our invitation (6 of whom had been involved in the development of QUADAS-2) formed the QUADAS-C advisory group. Together with the 8 members of the steering group, all 24 people participated in the Delphi study as panel members.

Before the first Delphi round, the steering group compiled an initial list of items that were considered potentially important for inclusion in the tool. The sources we consulted for identifying potentially important items included an overview of comparative DTA reviews published in 2017 (3), any risk-of-bias items associated with comparative DTA studies used in 102 Cochrane DTA review protocols with a comparative question (date of search in Cochrane Library: July 2018), and an article by Wade and colleagues, who described their experience in modifying QUADAS-2 for use in a comparative DTA systematic review (7). Only 1 meta-epidemiologic study provided empirical evidence of potential bias in comparative accuracy research (2). Studies investigating bias in randomized trials of interventions (15) were consulted as indirect evidence for items relating to the randomization process. The initial list of items was finalized during a face-to-face steering group meeting in September 2018 in Edinburgh, United Kingdom. This list, containing 16 items, fed into the first Delphi round. Details on the item-generating process are available in **Supplement 1** (available at [Annals.org](https://annals.org)).

The aims of Delphi rounds 1, 2, and 3 were to collect panel members' opinions regarding the fundamental properties and scope of QUADAS-C, to determine which items to include in the tool, and to generate additional items. Items were included in the tool or excluded from a Delphi round following a predefined threshold for consensus (70% agreement). Items not reaching this threshold were rerated in subsequent rounds with occasional amendments to wording. After round 3, the steering

**Table 1.** Differences Between Single-Test Accuracy and Comparative Accuracy Questions\*

| Feature                     | Accuracy of a Single Test                                                                                                            | Comparative Accuracy                                                                                                                                                                                                                                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Health-related question     | How accurately can an index test classify persons who have or do not have the target condition?                                      | How does the accuracy of index test A compare with that of index test B?                                                                                                                                                                                                                                                        |
| Ideal study design          | A study in which participants are consecutively or randomly sampled and all undergo a single index test and the reference standard   | A study in which participants are consecutively or randomly sampled and:<br>Each participant undergoes all index tests and the reference standard (fully paired or within-subject design)<br>or<br>Participants are randomly allocated to an index test and all participants receive the reference standard (randomized design) |
| Summary measures            | Sensitivity and specificity, predictive values, or other accuracy measures                                                           | Absolute or relative difference in sensitivity and specificity, predictive values, or other accuracy measures                                                                                                                                                                                                                   |
| Relevant for which purposes | Knowing the probability of disease after a test result<br>Finding the most appropriate position for a test in the diagnostic pathway | Estimating the change in accuracy when an alternative test is used<br>Informing decisions on which tests to use†                                                                                                                                                                                                                |

\* Adapted from reference 3.

† Factors other than comparative accuracy may inform decisions regarding test selection.

**Figure 1.** Proposed use of QUADAS-2 and QUADAS-C together.

| Domain 1: Patient Selection     |                                                                                                                           |                                 |                    |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------|
| Single-Test Accuracy (QUADAS-2) |                                                                                                                           | Answers for Test A              | Answers for Test B |
| Signaling questions             | 1.1 Was a consecutive or random sample of patients enrolled?                                                              | Yes/No/Unclear                  | Yes/No/Unclear     |
|                                 | 1.2 Was a case-control design avoided?                                                                                    | Yes/No/Unclear                  | Yes/No/Unclear     |
|                                 | 1.3 Did the study avoid inappropriate exclusions?                                                                         | Yes/No/Unclear                  | Yes/No/Unclear     |
| Risk of bias                    | 1.4 Could the selection of patients have introduced bias?                                                                 | Low/High/Unclear                | Low/High/Unclear   |
| Applicability concerns          | 1.5 Are there concerns that the included patients do not match the review question?                                       | Low/High/Unclear                | Low/High/Unclear   |
| Comparative Accuracy (QUADAS-C) |                                                                                                                           | Answers for the Test Comparison |                    |
| Signaling questions             | C1.1 Was the risk of bias for each index test judged "low" for this domain?*                                              | Yes/No                          |                    |
|                                 | C1.2 Was a fully paired or randomized design used?                                                                        | Yes/No/Unclear                  |                    |
|                                 | C1.3 Was the allocation sequence random?†                                                                                 | Yes/No/Unclear/Not applicable   |                    |
|                                 | C1.4 Was the allocation sequence concealed until patients were enrolled and assigned to index tests?†                     | Yes/No/Unclear/Not applicable   |                    |
| Risk of bias                    | C1.5 Could the selection of patients have introduced bias in the comparison?                                              | Low/High/Unclear                |                    |
| Domain 2: Index Test            |                                                                                                                           |                                 |                    |
| Single-Test Accuracy (QUADAS-2) |                                                                                                                           | Answers for Test A              | Answers for Test B |
| Signaling questions             | 2.1 Were the index test results interpreted without knowledge of the results of the reference standard?                   | Yes/No/Unclear                  | Yes/No/Unclear     |
|                                 | 2.2 If a threshold was used, was it prespecified?                                                                         | Yes/No/Unclear                  | Yes/No/Unclear     |
| Risk of bias                    | 2.3 Could the conduct or interpretation of the index test have introduced bias?                                           | Low/High/Unclear                | Low/High/Unclear   |
| Applicability concerns          | 2.4 Are there concerns that the index test, its conduct, or its interpretation differ from the review question?           | Low/High/Unclear                | Low/High/Unclear   |
| Comparative Accuracy (QUADAS-C) |                                                                                                                           | Answers for the Test Comparison |                    |
| Signaling questions             | C2.1 Was the risk of bias for each index test judged "low" for this domain?*                                              | Yes/No                          |                    |
|                                 | C2.2 Were the index test results interpreted without knowledge of the results of the other index test(s)?‡                | Yes/No/Unclear/Not applicable   |                    |
|                                 | C2.3 Is undergoing one index test unlikely to affect the performance of the other index test(s)?‡                         | Yes/No/Unclear/Not applicable   |                    |
|                                 | C2.4 Were the index tests conducted and interpreted without advantaging one of the tests?                                 | Yes/No/Unclear                  |                    |
| Risk of bias                    | C2.5 Could the conduct or interpretation of the index tests have introduced bias in the comparison?                       | Low/High/Unclear                |                    |
| Domain 3: Reference Standard    |                                                                                                                           |                                 |                    |
| Single-Test Accuracy (QUADAS-2) |                                                                                                                           | Answers for Test A              | Answers for Test B |
| Signaling questions             | 3.1 Is the reference standard likely to correctly classify the target condition?                                          | Yes/No/Unclear                  | Yes/No/Unclear     |
|                                 | 3.2 Were the reference standard results interpreted without knowledge of the results of the index test?                   | Yes/No/Unclear                  | Yes/No/Unclear     |
| Risk of bias                    | 3.3 Could the reference standard, its conduct, or its interpretation have introduced bias?                                | Low/High/Unclear                | Low/High/Unclear   |
| Applicability concerns          | 3.4 Are there concerns that the target condition as defined by the reference standard does not match the review question? | Low/High/Unclear                | Low/High/Unclear   |
| Comparative Accuracy (QUADAS-C) |                                                                                                                           | Answers for the Test Comparison |                    |
| Signaling questions             | C3.1 Was the risk of bias for each index test judged "low" for this domain?*                                              | Yes/No                          |                    |
|                                 | C3.2 Did the reference standard avoid incorporating any of the index tests?                                               | Yes/No/Unclear                  |                    |
| Risk of bias                    | C3.3 Could the reference standard, its conduct, or its interpretation have introduced bias in the comparison?             | Low/High/Unclear                |                    |
| Domain 4: Flow and Timing       |                                                                                                                           |                                 |                    |
| Single-Test Accuracy (QUADAS-2) |                                                                                                                           | Answers for Test A              | Answers for Test B |
| Signaling questions             | 4.1 Was there an appropriate interval between index tests and the reference standard?                                     | Yes/No/Unclear                  | Yes/No/Unclear     |
|                                 | 4.2 Did all patients receive a reference standard?                                                                        | Yes/No/Unclear                  | Yes/No/Unclear     |
|                                 | 4.3 Did all patients receive the same reference standard?                                                                 | Yes/No/Unclear                  | Yes/No/Unclear     |
|                                 | 4.4 Were all patients included in the analysis?                                                                           | Yes/No/Unclear                  | Yes/No/Unclear     |
| Risk of bias                    | 4.5 Could the patient flow have introduced bias?                                                                          | Low/High/Unclear                | Low/High/Unclear   |
| Comparative Accuracy (QUADAS-C) |                                                                                                                           | Answers for the Test Comparison |                    |
| Signaling questions             | C4.1 Was the risk of bias for each index test judged "low" for this domain?*                                              | Yes/No                          |                    |
|                                 | C4.2 Was there an appropriate interval between the index tests?                                                           | Yes/No/Unclear                  |                    |
|                                 | C4.3 Was the same reference standard used for all index tests?                                                            | Yes/No/Unclear                  |                    |
|                                 | C4.4 Are the proportions and reasons for missing data similar across index tests?                                         | Yes/No/Unclear                  |                    |
| Risk of bias                    | C4.5 Could the patient flow have introduced bias in the comparison?                                                       | Low/High/Unclear                |                    |

\* Refers back to the QUADAS-2 risk-of-bias judgments (questions 1.4, 2.3, 3.3, or 4.5).

† Only applicable to randomized designs.

‡ Only applicable if patients received multiple index tests (fully or partially paired designs).

group evaluated all 5 remaining items for which no consensus had been achieved and decided which items to include, providing justifications to the panel. In round 4, the proposed final list of included items was presented and panel members were invited to comment on the tool. The Delphi study led to the development of the first draft version of QUADAS-C, which was revised further in a face-to-face consensus meeting. The anonymized results of each Delphi round are available in **Supplement 1**.

### Consensus Meeting

We held a 2-day consensus meeting for the QUADAS-C group in August 2019 in Birmingham, United Kingdom, which was attended by 16 of 24 members (8 steering group members and 8 advisory group members). The main focus of the first day was to resolve remaining issues arising from the Delphi study through small-group discussions. Additionally, the group piloted the tool on 2 comparative DTA studies to identify challenges associated with its practical use. On the second day, the steering group critically reviewed the tool, discussed plans for piloting it, and agreed on the terminology to be used in the guidance document. Based on the outcomes of the meeting, the steering group revised QUADAS-C to its publicly pilotable version.

### Pilot Study

The last phase of the development was a pilot study to collect users' experiences with and feedback on using QUADAS-C (protocol registered at <https://osf.io/agx3z>). We recruited participants through various networks—some were authors of Cochrane Reviews, members of the Cochrane Screening and Diagnostic Tests Methods Group, or members of the GRADE (Grading of Recommendations Assessment, Development and Evaluation) Working Group. Other participants were recruited through our affiliated universities and from Twitter ([www.twitter.com](http://www.twitter.com)). Anyone interested in comparative DTA studies or systematic reviews, including health care providers, researchers, and students, was invited to pilot QUADAS-C on 1 of 4 comparative DTA studies purposely chosen to represent various designs (16–19). We also invited authors of ongoing systematic reviews to try out QUADAS-C in their review. Forty-four people participated in the pilot, of whom 6 piloted the tool in ongoing DTA systematic reviews (1 review [20] has been published) or other types of evidence syntheses. Results of the pilot study are available in **Supplement 1**. Although participants generally found the tool to be complete and easy to use, they also highlighted items that were ambiguous or in need of further explanation; this led us to make changes to item wording and to include brief explanations for each item in the tool. The steering group implemented these last changes and circulated the final version to the advisory group for approval.

### Role of the Funding Source

Amsterdam UMC (the Netherlands) provided funding for this study but had no role in the study design; collection, analysis, or interpretation of the data; or the decision to approve publication of the final manuscript.

## THE QUADAS-C TOOL

The final version of QUADAS-C can be found in **Supplement 2** (available at [Annals.org](http://Annals.org)) and at [www.quadas.org](http://www.quadas.org). The QUADAS-C tool is intended to assess the risk of bias of test comparisons undertaken in comparative DTA studies. It is designed to be an extension of QUADAS-2, meaning that it should be used together with QUADAS-2, because the risk-of-bias judgments from QUADAS-2 are required to make risk-of-bias judgments in QUADAS-C.

The QUADAS-C tool contains 14 signaling questions and 4 risk-of-bias judgment questions across the same 4 domains as QUADAS-2: Patient Selection, Index Test, Reference Standard, and Flow and Timing (**Figure 1**). In the remainder of this article, we will elaborate on the basic principles and structure of QUADAS-C. A more detailed explanation of how to use the tool is provided in the guidance document, which is available in **Supplement 3** (available at [Annals.org](http://Annals.org)) and at [www.quadas.org](http://www.quadas.org).

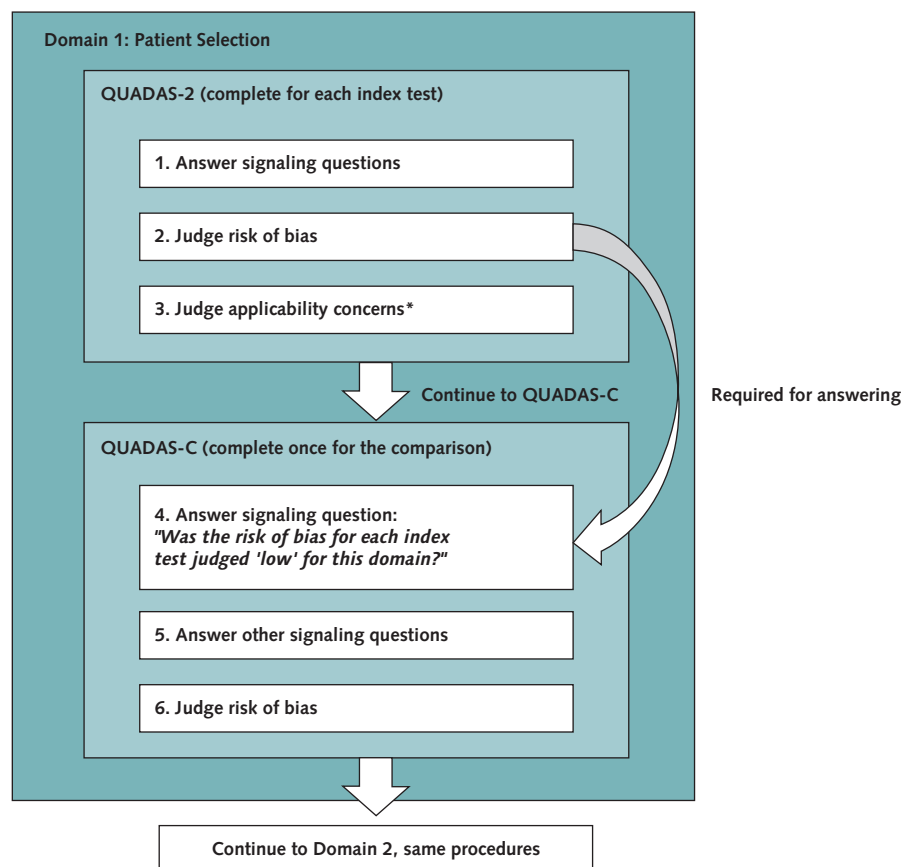
**Figure 1** provides our proposal on how to use the 2 tools together: QUADAS-2 is completed multiple times, once for each index test, whereas QUADAS-C is completed once per comparison. Additional columns can be added in QUADAS-2 for each additional test in the comparison.

### Scope of QUADAS-C

The QUADAS-C tool was designed primarily with assessment of fully paired and randomized comparative DTA studies in mind. Taken together, these comprise most comparative DTA study designs in systematic reviews (10). Although QUADAS-C could be used to assess other comparative DTA designs, it will need to be tailored to the specific design being assessed (for example, by including new signaling questions and removing irrelevant ones). In particular, in unpaired or partially paired studies without randomization, the issue of confounding will need to be addressed in more detail. The QUADAS-C tool is not designed to assess risk of bias in test comparisons made between studies (also called between-study or indirect comparisons).

Some comparative DTA studies may include a comparison between 1 or more testing strategies (that is, combinations of tests) to assess whether 1 testing strategy is more accurate than another test or testing strategy. The QUADAS-C tool can be used to assess these comparisons as well; however, users will need to define the comparison clearly, and careful tailoring of the tool may be required.

In contrast to QUADAS-2, QUADAS-C does not have questions on concerns regarding applicability. Users can nevertheless arrive at a judgment regarding applicability of the test comparison by choosing the highest concern (that is, the worst) applicability judgment for an index test in QUADAS-2. For example, an item in the Index Test domain of QUADAS-2 is, "Is there concern that the index test, its conduct, or interpretation differ from the review question?" If the answer for test A is "low concern" and the answer for test B is "high concern," it is clear that there is high concern regarding the applicability of the comparison between A and B. The assessment of

**Figure 2.** Process of using QUADAS-2 and QUADAS-C together.

\* With the exception of domain 4 (Flow and Timing).

applicability, although not part of QUADAS-C, is no less important than the assessment of risk of bias. Therefore, we strongly recommend that users also consider and describe the applicability of test comparisons, based on applicability judgments from QUADAS-2.

### How Is QUADAS-C Used Together With QUADAS-2?

Figure 2 shows schematically how QUADAS-2 and QUADAS-C are completed together when assessing a comparative DTA study. First, starting with the Patient Selection domain, QUADAS-2 is completed for each index test separately. Assuming a comparison of 2 tests, this will lead to risk-of-bias and applicability judgments for test A, followed by risk-of-bias and applicability judgments for test B.

Next, still within the Patient Selection domain, QUADAS-C is completed once for the comparison between tests A and B. The first signaling question of each domain in QUADAS-C, "Was the risk of bias for each index test judged 'low' for this domain?", makes use of the risk-of-bias judgments in QUADAS-2: If both risk-of-bias judgments for tests A and B were "low," this question is answered "yes," implying a low risk of bias for the comparison. By subsequently answering other QUADAS-C signaling questions in this domain, users can reach a risk-of-bias

judgment for the comparison. The same procedure is repeated for subsequent domains (Index Test, Reference Standard, Flow and Timing).

By having the signaling question "Was the risk of bias for each index test judged 'low' for this domain?" in each domain, QUADAS-C requires a low risk-of-bias judgment for each index test in QUADAS-2 for a low risk-of-bias judgment in QUADAS-C. When the risk of bias is "high" for 1 or more index tests in QUADAS-2, the potential for bias in the comparison exists. Although it may be possible that the direction and magnitude of bias affecting each index test may cancel each other out, such predictions are difficult to make.

### Assessing Risk of Bias With QUADAS-C

We recommend that users complete QUADAS-C in 4 phases, similar to the process for QUADAS-2: 1) clearly state the review question, 2) tailor the tool to each review and develop review-specific guidance, 3) review the study flow diagram or construct one if none is reported, and 4) judge risk of bias. The guidance document in Supplement 3 provides details of each phase. Whenever a study includes multiple comparisons of interest, QUADAS-C needs to be completed for each because a risk-of-bias judgment in QUADAS-C is specific to a particular test comparison.

**Information to Support the Judgment of Risk of Bias**

When judging risk of bias, users should record all of the information used to reach the judgment for reasons of transparency and reproducibility. For this purpose, QUADAS-C contains free text fields for recording the comparative study design (users can choose from a set of prespecified designs or describe the design) and information relevant to the validity of the comparison. The latter should be recorded for each of the 4 domains—for instance, how participants were allocated to index tests (Patient Selection domain) and whether there were any differences in the reasons for missing data between index tests (Flow and Timing domain).

**Answering Signaling Questions**

Signaling questions in QUADAS-C can be answered “yes,” “no,” or “unclear,” with “yes” indicating low risk of bias. A “no” answer implies that the potential for bias exists, but it does not automatically lead to a high risk-of-bias judgment for that domain; instead, users need to consider the likelihood and importance of the bias (see the next section). The options “yes” and “no” should also be used when the user’s assessment is “probably yes” or “probably no,” respectively. The option “unclear” is only appropriate if there is insufficient information to answer either “yes” or “no.” Detailed explanations with examples for answering each signaling question are provided in the guidance document (Supplement 3).

**Judging the Risk of Bias for Each Domain**

The answers to signaling questions will help the user arrive at a risk-of-bias judgment for each domain, which can be “low,” “high,” or “unclear.” A “yes” answer to all signaling questions within a domain should typically lead to a low risk-of-bias judgment. A “no” answer to a single signaling question may lead to a high risk-of-bias judgment if the associated bias is of such concern that the entire domain is deemed problematic; this is often a judgment call on the users’ part. Users may judge risk of bias as “unclear” if there is insufficient information to judge risk as either low or high.

**Judging the Overall Risk of Bias Across All Domains**

Although not formally part of QUADAS-C, users may find it helpful to produce an overall risk-of-bias judgment across all domains for each study. Examples would be to

judge “low overall risk of bias” if all domains were at low risk of bias and to judge “high overall risk of bias” or “unclear overall risk of bias” if 1 or more domains were at high or unclear risk of bias, respectively.

**Incorporating QUADAS-C Assessments in Comparative DTA Systematic Reviews**

**Narrative and Visual Summaries of Risk-of-Bias Judgments**

Users of QUADAS-C are strongly encouraged to provide a narrative and/or visual summary of their risk-of-bias judgments across studies. Table 2 shows an example of how to present QUADAS-2 and QUADAS-C results together. If the comparison is between 2 index tests, the combined use of QUADAS-2 and QUADAS-C will result in judgments for the accuracy of test A, the accuracy of test B, and the comparison between A and B. If the review question only concerns comparative accuracy, users may decide to display only QUADAS-C results. The guidance document (Supplement 3) contains additional suggestions on how to present results.

**Using Risk-of-Bias Judgments to Inform the Analysis, Conclusions, and the Certainty of Evidence**

Risk-of-bias judgments can be used to investigate between-study heterogeneity (either by subgroup analysis or metaregression) or to explore the effect of excluding particular studies from meta-analyses (21). Such analyses can be done using risk-of-bias judgments for a particular domain or overall risk-of-bias judgments across domains. For example, users may assess whether studies at high risk of bias show different relative accuracy compared with studies at low risk of bias. Users may decide to exclude studies at high risk of bias from the primary analysis or as a sensitivity analysis. Ideally, QUADAS-C results should also be incorporated in the conclusions of systematic reviews (22). Risk-of-bias judgments can further inform assessments of the certainty, quality, or strength of the overall body of evidence (23).

**DISCUSSION**

Decisions regarding the selection of diagnostic tests for clinical practice may benefit from trustworthy evidence

**Table 2.** Suggestion of How to Present QUADAS-2 and QUADAS-C Results Together\*

| Study        | Test | Risk of Bias (QUADAS-2) |   |   |    | Applicability Concerns (QUADAS-2) |   |   | Risk of Bias (QUADAS-C) |   |   |    |
|--------------|------|-------------------------|---|---|----|-----------------------------------|---|---|-------------------------|---|---|----|
|              |      | P                       | I | R | FT | P                                 | I | R | P                       | I | R | FT |
| Author, year | A    | ✓                       | X | ✓ | ✓  | ✓                                 | ✓ | ✓ | X                       | X | ✓ | ✓  |
|              | B    | ✓                       | ✓ | ✓ | ✓  | ✓                                 | ✓ | X |                         |   |   |    |
| Author, year | A    | ?                       | ✓ | ✓ | X  | ✓                                 | ? | ✓ | ?                       | X | ✓ | X  |
|              | B    | ?                       | ✓ | ✓ | X  | ✓                                 | ✓ | ✓ |                         |   |   |    |
| Author, year | A    | ✓                       | ✓ | ✓ | ✓  | ?                                 | ✓ | ✓ | ✓                       | ? | ? | ✓  |
|              | B    | ✓                       | ? | ✓ | ✓  | ?                                 | ✓ | ✓ |                         |   |   |    |

FT = Flow and Timing; I = Index Test; P = Patient Selection; R = Reference Standard; ✓ = low risk; X = high risk; ? = unclear risk.  
 \* This table may be simplified if the QUADAS-2 judgments for P, R, and FT are the same for each index test. See Supplement 3 for this and other examples on how to present results. Templates for tabular and graphical presentations are available at [www.quadas.org](http://www.quadas.org).

on the relative accuracy of alternative tests. Although comparative DTA studies can provide valid evidence on relative test performance, it is essential that such studies are critically evaluated for any shortcomings in their design, conduct, and analysis that may bias their results.

We developed QUADAS-C through a rigorous process of iterative feedback, consensus, and user testing. The QUADAS-C tool is explicitly developed with the structure and design of QUADAS-2 in mind, so that users who have experience with QUADAS-2 may find the extension straightforward to use. We acknowledge that the items in QUADAS-C are mainly based on consensus and theoretical considerations; empirical confirmation of bias is still limited. Like many quality assessment tools, we expect that QUADAS-C will need updating as knowledge regarding biases in comparative DTA studies evolves over time.

The QUADAS-C tool has been designed as a generic tool for comparing all types of diagnostic tests. Because unique methodological considerations may apply to specific diagnostic tests, users are invited to tailor the tool to the individual systematic review by adding, omitting, or modifying signaling questions. For example, PROBAST (Prediction model Risk Of Bias ASsessment Tool) (24) provides more specific signaling questions for multivariable models that users could consider when tailoring.

It should be noted that QUADAS-C is not appropriate for assessing the risk of bias in studies that evaluate the effectiveness of test-treatment strategies on people-important outcomes, such as morbidity and mortality. For those studies, users should use tools matching the type of study, such as the revised Cochrane risk-of-bias tool for randomized trials (25) and ROBINS-I (Risk Of Bias In Non-randomised Studies - of Interventions) for non-randomized studies of interventions (26).

As observed during the Delphi rounds and the pilot study, the use of QUADAS-C is not without challenges. Because the tool is used together with QUADAS-2, users (especially those unfamiliar with QUADAS-2) may find the combined number of signaling questions to be large. Furthermore, assessing the risk of bias in test comparisons with 3 or more tests, although possible, may be challenging. The QUADAS-C tool was designed with fully paired and randomized studies in mind, and its use for assessing nonrandomized and other “creative” designs will require additional tailoring of the tool. The development of a web-based tool, which is currently planned, may resolve some of the issues raised by users, such as automated completion of conditional signaling questions, optional display of explanations to signaling questions, and automated construction of exportable risk-of-bias tables and graphs that combine QUADAS-2 and QUADAS-C results.

We hope that QUADAS-C will help review authors systematically perform risk-of-bias assessments and identify high-quality studies in comparative DTA systematic reviews; help primary study investigators avoid potential biases in the design and conduct of their study; and, more generally, increase awareness of the importance of methodological rigor among those involved in comparative

accuracy research. The tool may also raise awareness that comparing test accuracy using estimates obtained from noncomparative studies, a common practice in systematic reviews (3), is intrinsically at risk of generating biased results, reinforcing the need for well-designed comparative DTA studies to inform decision making regarding preferred tests.

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**Reproducible Research Statement:** *Study protocol:* Protocol of the Delphi study is available at <https://osf.io/tmze9>; protocol of the pilot study is available at <https://osf.io/agx3z>. *Statistical code:* Not applicable. *Data set:* Not available.

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